

HIGH-YIELD CHEMISTRY MOCK TEST (SET 1)

Based on PYQ Sets 56/5/1, 56/7/1, and HFG1E/5

Time Allowed: 3 Hours

Maximum Marks: 70

By Pragya

General Instructions:

- **Section A:** 10 Multiple Choice Questions (Highly repetitive concepts).
- **Section B:** Very Short Answer (2 Marks).
- **Section C:** Short Answer (3 Marks).
- **Section D:** Case-Based Questions (4 Marks).
- **Section E:** Long Answer (5 Marks).

SECTION A (1 Mark Each)

1. Which of the following solutions will have the highest boiling point?

- (A) 1.0 M NaOH
- (B) 1.0 M Na₂SO₄
- (C) 1.0 M NH₄NO₃
- (D) 1.0 M KNO₃

Concept: Van't Hoff factor i & Colligative Properties - Repeated in all sets

2. Which of the following transition metal ions is colourless?

- (A) Ti³⁺
- (B) Sc³⁺
- (C) Mn²⁺
- (D) Fe³⁺

Concept: d-d transition & unpaired electrons - Repeated in all sets

3. The specific sequence in which amino acids are arranged in a protein is called its:

- (A) Primary structure
- (B) Secondary structure
- (C) Tertiary structure
- (D) Quaternary structure

Concept: Protein structure - Repeated in 56/5/1 & HFG1E/5

4. Identify the reaction: $\text{Ar}-\text{N}_2^+ \text{X}^- \longrightarrow \text{Ar}-\text{CN} + \text{N}_2$ (Reagent: CuCN/KCN)

- (A) Sandmeyer's reaction
- (B) Gattermann reaction
- (C) Friedel-Crafts reaction
- (D) Stephen reaction

Concept: Sandmeyer's Reaction - Repeated in 56/5/1 & HFG1E/5

5. For a zero-order reaction, the slope of the plot of concentration of reactant vs time is:

- (A) $-k$
- (B) k
- (C) $k/2.303$
- (D) $-k/2.303$

Concept: Kinetics Graphs - Repeated in 56/5/1 & 56/7/1

SECTION B (2 Marks Each)

6. (a) State Kohlrausch's law of independent migration of ions.

(b) How does the conductivity of a solution change with dilution?

Source: Repeated in 56-5-1, 56-7-1, HFG1E/5

7. Account for the following:

- (i) The dipole moment of chlorobenzene is lower than that of cyclohexyl chloride.
- (ii) Alkyl halides, though polar, are immiscible with water.

Source: Repeated reasoning topics in Organic sections

8. Biomolecules:

Write the chemical reaction of D-Glucose with **HCN** and **Bromine water**. What does each reaction indicate about the structure of glucose?

Source: Repeated in 56-7-1 & HFG1E/5

9. Solutions:

Define **Osmotic Pressure**. Why is it preferred for determining the molar mass of macromolecules like proteins?

Source: Repeated concept in 56-7-1 & HFG1E/5

SECTION C (3 Marks Each)

10. Chemical Kinetics (Numerical):

A first-order reaction takes 40 minutes for 30% decomposition. Calculate the half-life ($t_{1/2}$) for this reaction.

OR

Show that for a first-order reaction, the time required for **99% completion is twice** the time required for the completion of **90%** of the reaction.

Source: "99% vs 90%" proof appears in 56-5-1 and HFG1E/5

11. Coordination Compounds:

- (i) Write the IUPAC name of $[\text{Co}(\text{NH}_3)_5(\text{ONO})]\text{Cl}_2$.
- (ii) Using Valence Bond Theory (VBT), explain the geometry and magnetic behavior of $[\text{Ni}(\text{CN})_4]^{2-}$. (Atomic No. Ni = 28).

Source: VBT and IUPAC naming appear in all three papers

12. Electrochemistry (Numerical):

Calculate the emf of the following cell at 298 K:
 $\text{Mg(s)} \mid \text{Mg}^{2+}(0.001 \text{ M}) \parallel \text{Cu}^{2+}(0.0001 \text{ M}) \mid \text{Cu(s)}$
(Given: $E^\circ_{\text{Mg}^{2+}/\text{Mg}} = -2.37 \text{ V}$, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34 \text{ V}$).

Source: Nernst Equation numerical is guaranteed in all sets

13. Organic Mechanism:

- (i) Which alkyl halide from the following pair would you expect to react more rapidly by an $\text{S}_{\text{N}}1$ mechanism? Explain.
 $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--Br}$ OR $(\text{CH}_3)_3\text{C--Br}$
- (ii) Racemisation occurs in $\text{S}_{\text{N}}1$ reactions. Why?

Source: $\text{S}_{\text{N}}1/\text{S}_{\text{N}}2$ comparison appears in all three papers

SECTION D (4 Marks)

14. Read the passage and answer the questions:

Topic: Crystal Field Theory (CFT) & Colour

The colour of coordination compounds depends on the crystal field splitting (Δ_o). When ligands approach the central metal ion, the d-orbitals split into two sets (t_{2g} and e_g). Excitation of electrons between these sets absorbs specific wavelengths of light.

- (a) Explain why $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ is violet in colour but becomes colourless when heated to remove water. (2M)
- (b) What is the relationship between the strength of the ligand and the crystal field splitting energy (Δ_o)? (1M)

(c) Compare the crystal field splitting energy (Δ_o) for the complexes $[\text{Fe}(\text{CN})_6]^{4-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$. (1M)

Source: Based on Case Study in 56-7-1 and 56-5-1

SECTION E (5 Marks Each)

15. (a) d- and f-Block Elements:

(i) Account for the following:

- Transition metals and their compounds act as good catalysts.
- E° value for $\text{Mn}^{3+}/\text{Mn}^{2+}$ couple is highly positive compared to $\text{Cr}^{3+}/\text{Cr}^{2+}$.
- Zr and Hf have almost identical atomic radii.

(ii) What happens when acidified Potassium Dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) reacts with Potassium Iodide (KI)? Write the ionic equation.

Source: Trends and Lanthanoid contraction repeat in all sets

16. (a) Aldehydes, Ketones, and Carboxylic Acids:

(i) An organic compound (A) with molecular formula $\text{C}_8\text{H}_8\text{O}$ forms an orange-red precipitate with 2,4-DNP reagent and gives a yellow precipitate on heating with iodine and NaOH. It does not reduce Tollen's reagent. On drastic oxidation with chromic acid, it gives a carboxylic acid (B) having molecular formula $\text{C}_7\text{H}_6\text{O}_2$. Identify (A) and (B) and write the reactions involved.

(ii) Write the chemical equation for the **Cannizzaro reaction**.

OR

(b) Distinction Tests & Reactivity:

(i) Give a simple chemical test to distinguish between:

- Propanal and Propanone.
- Phenol and Benzoic acid.

(ii) Arrange the following in increasing order of their boiling points:

CH_3CHO , $\text{CH}_3\text{CH}_2\text{OH}$, CH_3OCH_3 , $\text{CH}_3\text{CH}_2\text{CH}_3$

(iii) Why are carboxylic acids more acidic than phenols?

Source: Cannizzaro and Iodoform tests are the most repeated organic topics.